Dasturchi va administrator qo‘llanmasi

O‘rnatish, sozlash, exam.ajou.uz da joylashtirish, boshqaruv, texnik xizmat, zaxira va tiklash, unumdorlik, xavfsizlik va muammolarni hal qilish

Versiya 1.3 · 2026-yil oktabr

Ajou University in Tashkent · AUT Secure Exam

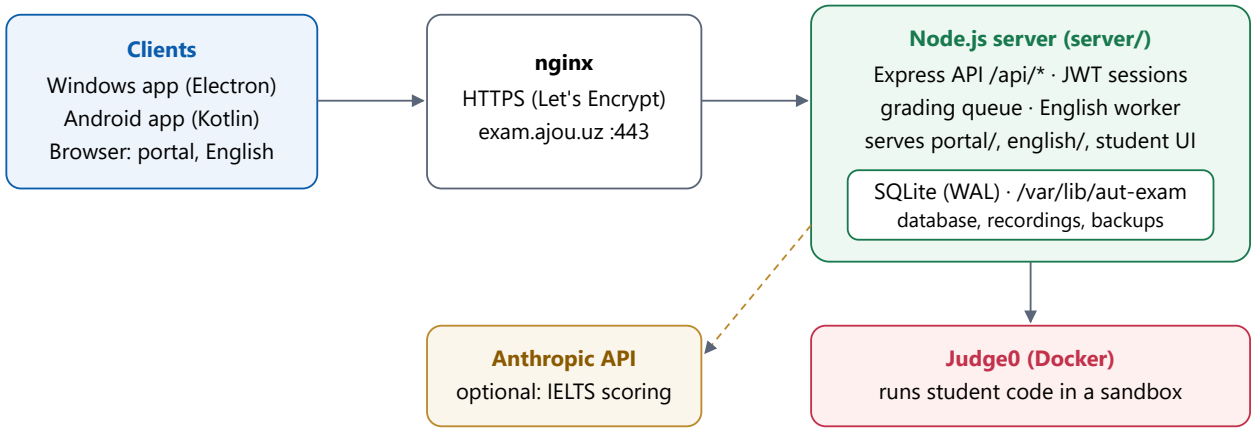
> Tizimni ishlatadigan dasturchilar va IT xodimlari uchun. O‘lchangan quvvat va barcha sinov dalillari alohida **Loyiha holati va sinov hisoboti**da; serverni bosqichma-bosqich o‘rnatish **AUT_Secure_Exam_Deployment_Guide.pdf** da.

Mundarija

1. Tizim haqida umumiy ma’lumot
2. Ishlab chiqish muhiti
3. Sozlamalar (.env)
4. exam.ajou.uz da joylashtirish
5. Rollar va boshqaruv
6. Ilovalarni yig‘ish va chiqarish
7. Texnik xizmat va kuzatuv
8. Zaxira va tiklash
9. Unumdorlik va kengaytirish
10. Xavfsizlik
11. Sinovlar
12. Muammolarni hal qilish

1. Tizim haqida umumiy ma’lumot

Bitta Node.js server hamma narsani saqlaydi: hisoblar, imtihonlar, javoblar, baholar, ingliz tili amaliyoti. Barcha mijozlar uning HTTPS API sidan foydalanadi. Dasturlash javoblarini alohida mashinadagi Judge0 himoyalangan muhitda ishga tushiradi; ishchi rejimda server talaba kodini hech qachon o‘zi ishga tushirmaydi.



Tarkibiy qismlar va ularning o‘zaro aloqasi

Papka	
`server/`	API, ma’lumotlar bazasi, baholash, ingliz tili, sinovlar, yuklama va avariya sinovlari
`portal/`	Professor va administrator portali
`english/`	Ingliz tili veb-ilovasi
`desktop/`	Windows ilovasi (Electron); renderer/ /student/ manzilida ham ochiladi
`android/`	Android ilovasi (Kotlin, Jetpack Compose)
`judge0/`	Judge0 himoyalangan muhiti (Docker Compose)
`deploy/`	O‘rnatuvchi, nginx, systemd, zaxira taymeri, ishchi .env
`docs/`	Qo‘llanmalar

2. Ishlab chiqish muhiti

Talablar: Node.js 22 LTS, Git; Android ilovasi uchun Android Studio; Judge0 uchun Docker + Linux (cgroup v1). Judge0 bo‘lmasa, ishlab chiqishda `GRADER=local` (kodni kompyuteringizda ishga tushiradi, himoyasiz) yoki `GRADER=off` ishlatiladi.

```
# Windows or Linux, Node.js 22 LTS
git clone https://github.com/ramzanali-blip/AJOU-EXAM.git
cd AJOU-EXAM/server
npm ci
copy .env.example .env          (Linux: cp .env.example .env)
npm run seed                    # demo accounts, development only
npm start                       # http://localhost:4000
npm test                        # automated tests
```

3. Sozlamalar (.env)

Sozlama	Misol	Ma‘nosi
`NODE_ENV`	`production`	Zaif sozlamalarni rad etadi (qisqa JWT_SECRET, GRADER=local).
`JWT_SECRET`	`96 hex chars`	Seanslarni imzolaydi. O‘rnatuvchi yaratadi. O‘zgartirilsa, hamma tizimdan chiqadi.
`JWT_EXPIRES_IN`	`12h`	Seans davomiyligi.
`DB_PATH`	`/var/lib/aut-exam/exam.db`	SQLite ma‘lumotlar bazasi fayli.
`PROFESSOR_SIGNUP_CODE`	`(empty)`	Bo‘sh = professorlarni faqat administrator yaratadi.
`GRADER`	`judge0 | off`	judge0 = himoyalangan kod baholash; off = kodni qo‘lda baholash (ochiq sinov serverlari).
`JUDGE0_URL / JUDGE0_AUTH_TOKEN`	`http://10.x.x.x:2358`	Judge0 manzili (yopiq tarmoq) va tokeni.
`GRADER_CONCURRENCY`	`4`	Bir vaqtda baholanadigan urinishlar.
`AUTH_RATE_LIMIT`	`50`	Har bir IP uchun 15 daqiqada kirishlar. Butun guruh bitta kampus IP sidan chiqsa, oshiring (masalan, 1000).
`UV_THREADPOOL_SIZE`	`8 (systemd)`	Kirish ko‘p bo‘lganda parol xeshlash uchun ishchi oqimlar.
`ANTHROPIC_API_KEY`	`(optional)`	Ingliz tili speaking/writing uchun AI examiner.
`ENGLISH_ADMIN_EMAILS`	`dev@ajou.uz`	Ingliz tili topshiriqlari kutubxonasi muharrirlari.
`ENGLISH_CONCURRENCY`	`2`	Bir vaqtda matnga aylantiriladigan speaking javoblari (CPU ko‘p talab qiladi).

4. exam.ajou.uz da joylashtirish

Universitet sayti www.ajou.uz o‘zgarmaydi: tizim o‘z subdomeniga ega bo‘ladi. To‘liq ko‘rsatmalar, server o‘lchamlari va ishga tushirish ro‘yxati: **AUT_Secure_Exam_Deployment_Guide.pdf**. Qisqacha:

```
# 1. DNS (university IT): A record  exam ->
# 2. On the Ubuntu 22.04/24.04 server, from the source folder:
sudo DOMAIN=exam.ajou.uz ADMIN_EMAIL=it@ajou.uz bash deploy/install-ubuntu.sh
# 3. Judge0 (see judge0/README.md), then in /opt/aut-exam/server/.env:
#    JUDGE0_URL=http://:2358   JUDGE0_AUTH_TOKEN=
sudo aut-exam restart
# 4. First administrator (a professor account with admin rights)
sudo aut-exam create-professor "Full Name" name@ajou.uz 'strong-password' --admin
sudo aut-exam health
```

5. Rollar va boshqaruv

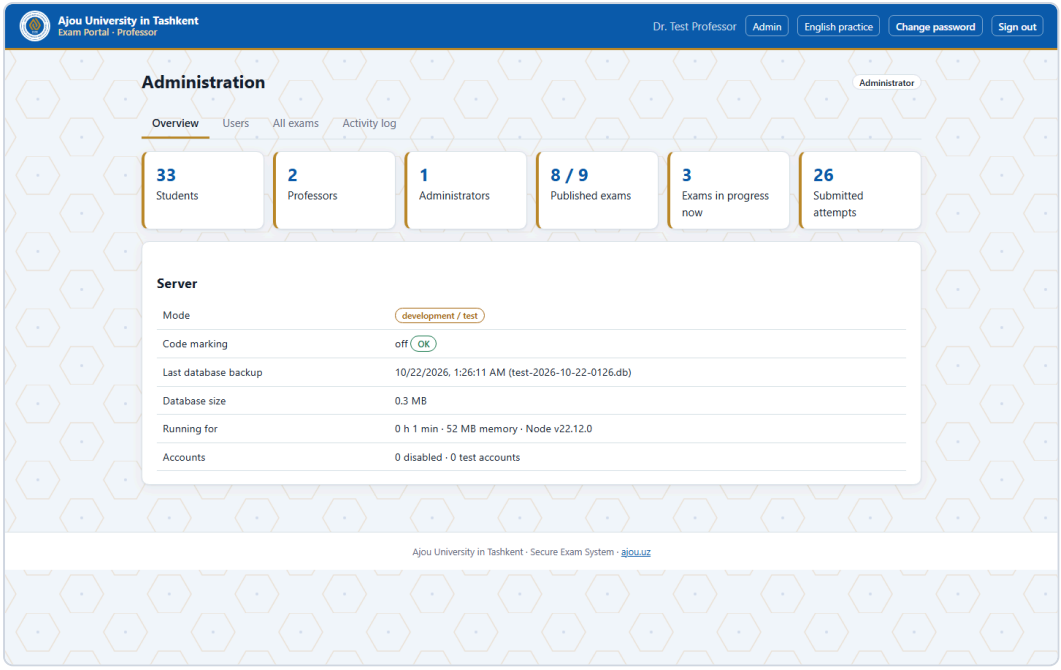
Uchta rol bor. Administratorlar — administrator huquqiga ega professor hisoblari. Server har bir so‘rovda rol va egalikni tekshiradi (professor faqat o‘z imtihonlarini, talaba faqat o‘z urinishlarini ko‘radi).

Harakat	Talaba	Professor	Admin
Kirish, imtihonga qo‘shilish, topshirish, e’lon qilingan natijalar, ingliz tili	✓	✓	✓
Imtihon yaratish va rejalashtirish, savollar, talabalarni yozish, monitoring, baholash, natijalar	–	✓	✓
O‘z imtihonidagi talabalar parolini tiklash	–	✓	✓
Barcha hisoblarni boshqarish (o‘chirish, parol tiklash, rolni o‘zgartirish, admin huquqi)	–	–	✓
Barcha imtihonlar, faoliyat jurnali, server holati va oxirgi zaxira	–	–	✓

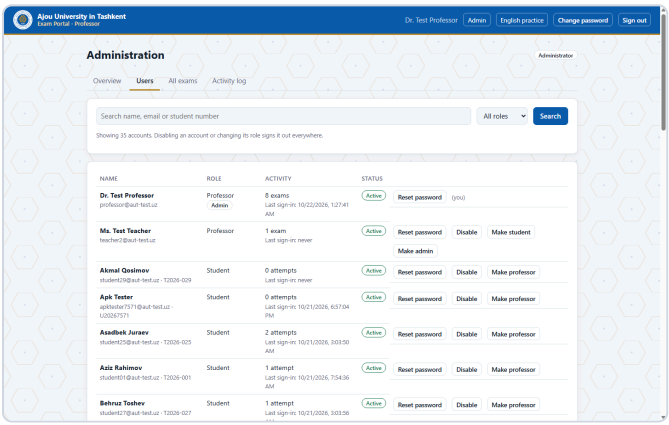
Serverda hisoblar va administratorlar yaratish:

```
sudo aut-exam create-professor "Full Name" name@ajou.uz 'password'          # professor
sudo aut-exam create-professor "Full Name" name@ajou.uz 'password' --admin  # administrator
cd /opt/aut-exam/server && sudo -u aut-exam npm run make-admin -- name@ajou.uz [--remove]
sudo aut-exam remove-test-accounts --yes      # before real exams
```

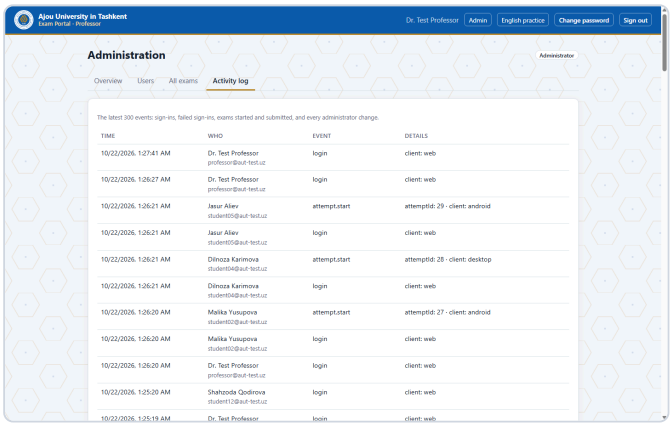
Portalda administratorlar yuqori panelda **Admin** ni ko‘radi: server holati va oxirgi zaxira, barcha foydalanuvchilar (qidirish, o‘chirish, parolni tiklash, rolni o‘zgartirish, admin huquqi), barcha imtihonlar hamda kirishlar, muvaffaqiyatsiz kirishlar va har bir administrator o‘zgarishi jurnali.



Admin → Overview: foydalanuvchilar, davom etayotgan imtihonlar, kod baholash, oxirgi zaxira



Admin → Users



Admin → Activity log

6. Ilovalarni yig‘ish va chiqarish

```
cd android
gradlew assembleStandardRelease bundleStandardRelease     # signed APK/AAB for https://exam.ajou.uz
gradlew assembleStandardRelease -PserverUrl=https://other.ajou.uz
gradlew assembleTrialDebug -PtestServerUrl=http://:4200   # "AUT Exam TEST"
cd desktop
npm ci && npm run dist                                     # Windows installer (examServerUrl in
package.json)
```

Imzo kalitlari: `android/aut-release.jks` + `android/keystore.properties` (hech qachon git ga emas; zaxiralang). Har bir Android chiqarilishida `versionCode` / `versionName` ni, desktop/package.json da `version` ni oshiring.

7. Texnik xizmat va kuzatuv

Xizmat ishdan chiqqanda yoki qayta yuklanganda avtomatik qayta ishga tushadi. Imtihonlar davrida Admin overview ni har kuni tekshiring.

```
sudo aut-exam status      # service state
sudo aut-exam logs        # live log (journald)
sudo aut-exam health      # {"ok":true,"grader":"judge0",...}
sudo aut-exam restart     # graceful: requests in flight are finished
sudo aut-exam backup      # backup now
# update: copy the new source, then run the installer again (data and .env are kept)
sudo DOMAIN=exam.ajou.uz ADMIN_EMAIL=it@ajou.uz bash deploy/install-ubuntu.sh
```

8. Zaxira va tiklash

Avtomatik: har soatda (systemd taymeri), eng yangi 60 nusxa `/var/lib/aut-exam/backups` da, SQLite `VACUUM INTO` bilan yaratiladi — talabalar yozayotgan paytda ham xavfsiz. Ularni har kecha boshqa mashinaga ko‘chiring:

```
# /etc/cron.d/aut-exam-offsite: copy backups and recordings to another machine every night at 02:30
30 2 * * * root rsync -a /var/lib/aut-exam/backups /var/lib/aut-exam/english-audio backup@backup-
host:/srv/aut-exam/
```

Tiklash (`npm run crash-test` bilan sinovdan o‘tgan):

```
sudo systemctl stop aut-exam
sudo cp /var/lib/aut-exam/backups/exam-YYYY-MM-DD-HHMM.db /var/lib/aut-exam/exam.db
sudo rm -f /var/lib/aut-exam/exam.db-wal /var/lib/aut-exam/exam.db-shm
sudo chown aut-exam:aut-exam /var/lib/aut-exam/exam.db
sudo systemctl start aut-exam && sudo aut-exam health
```

> Har oy nusxadan tiklashni sinab ko‘ring. Xuddi shu diskdagi zaxira disk buzilsa saqlanib qolmaydi.

9. Unumdorlik va kengaytirish

Tuzilishi: bitta Node.js jarayoni; WAL rejimidagi SQLite (har bir javobni saqlash bitta qisqa tranzaksiya); parol xeshlash (scrypt) ishchi oqimlarda; baholash fon navbatida; seanslar holatsiz tokenlar. O'lchangan natijalar sinov hisobotida. Boshqaruv vositalari:

Tor joy	Nima qilish kerak
Kirish ko'pligi	Talabalar imtihondan 10–15 daqiqa oldin kirsin (seans 12 soat amal qiladi). Ko'proq CPU yadrolari + UV_THREADPOOL_SIZE.
Kodni baholash	Ko'proq Judge0 ishchilari / kattaroq Judge0 VM; GRADER_CONCURRENCY.
Ingliz tili speaking	ENGLISH_CONCURRENCY = har bir CPU yadrosiga 1–2; ko'proq yadro.
Server o'lchami	Bir vaqtda ~1000 talabagacha: 4 vCPU / 8 GB; ko'proq: 8 vCPU / 16 GB (serverda load-test bilan tekshiring).
Bitta serverdan ko'p	Ko'p imtihonlarda bir vaqtda minglab talabalar: PostgreSQL va nginx ortida bir nechta server nusxasi (server/src/db.js sxemasi bevosita mos keladi).

10. Xavfsizlik

Soha	Qanday himoyalangan
Parollar	scrypt (N=16384), eski bcrypt xeshlari kirishda yangilanadi; kamida 8 belgi.
Kirish himoyasi	10 marta noto'g'ri parol hisobni 15 daqiqaga bloklaydi; IP bo'yicha cheklov (AUTH_RATE_LIMIT).
Seanslar	Imzolangan tokenlar (HS256, 12 soat). Parolni o'zgartirish/tiklash, o'chirish yoki rolni o'zgartirish barcha seanslarni darhol yakunlaydi.
Rollar	Har bir so'rovda tekshiriladi; imtihon va urinishlarga egalik tekshiruvi; administrator o'z huquqini olib tashlay olmaydi.
Uzatish	Faqat HTTPS (nginx + Let's Encrypt), HSTS; Android ilovasining ishchi versiyasi http ni rad etadi.
Imtihon halolligi	Serverdagi taymer va muddat; ilovalardagi himoya va hodisalar hisoboti; yuborilgandan keyin javoblar qulflanadi; javob kalitlari talabaga yuborilmaydi.
Kodni ishga tushirish	Faqat Judge0 da (alohida VM, yopiq tarmoq); ishchi rejimda GRADER=local rad etiladi.
Maxfiy ma'lumotlar	server/.env (640, root:aut-exam), judge0.conf, Android kaliti: hech qachon git yoki emailda emas.
Audit	Kirishlar, muvaffaqiyatsiz kirishlar, imtihon boshlanishi/yuborilishi va har bir administrator o'zgarishi qayd etiladi.
Kutubxonalar	npm audit: ma'lum zaifliklar 0 ta (2026-yil oktabr). Har chiqarilishdan oldin ishga tushiring.

☐ HTTPS ishlaydi, http yo'naltiriladi
☐ NODE_ENV=production, PROFESSOR_TEST_SIGNUP_CODE bo'sh
☐ Judge0 porti faqat imtihon serveridan ochiq

☐ SSH faqat kalit bilan; unattended-upgrades yoqilgan
☐ Sinov hisoblari o‘chirilgan; demo hisoblar o‘chirilgan
☐ Zaxiralar serverdan tashqariga ko‘chiriladi; tiklash sinalgan

11. Sinovlar

Barcha sinov vositalari o‘zining vaqtinchalik serveri va bazasini ishga tushiradi; haqiqiy ma’lumotlarga tegilmaydi.

```
cd server
npm test                          # 14 automated tests: exam flow, grading, English, security,
GRADER=off
npm run load-test                 # stages 100..2000 students (own throw-away server)
npm run load-test -- 500 1000     # chosen stages; PRELOGIN=1 = students signed in before the exam
npm run crash-test                # kills the server mid-exam, restarts, checks every answer; backup
+ restore
LOAD_URL=https://exam.ajou.uz LOAD_PROF_EMAIL=... LOAD_PROF_PASSWORD=... npm run load-test -- 200
# real server
```

Kompyuterdan ochiq sinov serveri (hisob shart emas): `Start Public Test Server.cmd` sinov serverini (`server/.env.test`, port 4200, `GRADER=off`) va Cloudflare tunnelini ishga tushiradi va ulashish uchun havolani ko‘rsatadi. O‘sha havoladagi yuklab olish sahifasi TEST ilovasini bitta bosish bilan ulaydi.

12. Muammolarni hal qilish

Muammo	Tekshirish / yechim
502 Bad Gateway	sudo aut-exam logs; ko‘pincha .env dagi xato yoki qisqa JWT_SECRET.
Yangilanishdan keyin xizmat ishga tushmayapti	node -v 22+ bo‘lishi kerak; o‘rnatuvchini qayta ishga tushiring; jurnalni tekshiring.
"Grader error" / Judge0 ga ulanib bo‘lmayapti	Judge0 VM da docker compose ps; JUDGE0_URL/token; firewall; cgroup v1. So‘ng Re-run auto-grading.
Ko‘p talaba bir vaqtda kira olmayapti	Jurnalda 429 ni qidiring: guruh bitta IP dan chiqsa, AUTH_RATE_LIMIT ni oshiring.
Talaba uchun "Account locked"	15 daqiqa kuting yoki parolni tiklang (Admin → Users).
Imtihon boshida sekin	Talabalar ertaroq kirsin; CPU ni tekshiring (top); yadro qo‘shing; 9-bo‘limga qarang.
Baza bloklangan / disk to‘lgan	df -h; joy bo‘shating; eski zaxiralar avtomatik o‘chiriladi (60 tasi saqlanadi).
Sertifikat muddati o‘tgan	sudo certbot renew; 80-port ochiqligini tekshiring.